DJK3C: DIGITAL ELECTRONICS

Unit I

Number System:

Decimal – binary – octal – hexadecimal number system – conversion from one system to another – binary arithmetic – 1's complement – 2's complement – BCD, excess 3, gray , alpha numeric codes.

Unit II

Boolean algebra:

Boolean operation – rules and laws of Boolean algebra – De Morgan's theorems – implication of expressions using Boolean algebra – Karnaugh map.

Unit III

Basic Logic gates:

AND, OR, NOT (symbol, truth table, circuit diagram, working) NAND, NOR, EX-OR, EX- NOR (symbol, truth table)

Unit IV

Combinational Circuits:

Half adder – full adder – half sub tractor – full sub tractor – binary adder – BCD adder – decoder – encoder – multiplexer – de multiplexer.

Unit V

Flip flops:

RS, JK, D, T flip flops – master slave flip flop - IC 555 timer – astable multi vibrator - mono stable multi vibrator.

Books for study and reference :

1. Digital principles & applications – Albert Paul Malvino & Leach
2. Digital Logic & Computer Design – Morris Mano.

Unit INumber System

Decimal - Binary- Octal - Hexadecimal number system – conversion from one system to another – Binary arithmetic – 1's complement – 2's complement – BCD, Excess, Gray, Alpha numeric codes

1.1 Number System:

A number is a mathematical object used to count, measure, and label. Numbers are represented by a string of digital symbols. A number system of base r is a system that uses distinct symbols for r digits. That is in a positional base$r$ numeral system r basic symbols (or digits) corresponding to the first r natural numbers including zero are used. To generate the rest of the numerals, the position of the symbol in the figure is used. The symbol in the last position has its own value, and as it moves to the left its value is multiplied by r. There are four systems of arithmetic used in digital system. These systems are Decimal, Binary, Hexadecimal and Octal.

System	Base	Digits
Binary	2	0 1
Octal	8	0 1 2 3 4 5 6 7
Decimal	10	0 1 2 3 4 5 6 7 8 9
Hexadecimal	16	0 1 2 3 4 5 6 7 8 9 A B C D E F

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1.2 Decimal Number System:

The Decimal number system has a base ten. This system uses ten distinct digits 0 1 2 3 4 5 6 7 8 9 to form any number. Each digit can be used individually or they can be grouped to form a numeric value.Each of decimal digits, 0 through 9, has a place value or weight depending on its position. The weights are units, tens, hundreds, thousands and so on. The same can be expressed as the powers of its base as $10^0, 10^1, 10^2, 10^3 \cdots etc$for the integer partand $10^{-1}, 10^{-2}, 10^{-3}, 10^{-4} \cdots etc$ for the fractional part. $10^0, 10^1, 10^2, 10^3 \cdots etc$represents the units, tens, hundreds, thousands etc. and the quantities $10^{-1}, 10^{-2}, 10^{-3}, \cdots etc$ represents one tenth, one hundredth, one thousandth etc. The integer part and fractional parts are separated by a decimal point. The position weights in decimal system is given as

$\cdots$	10^3	10^2	10^1	10^0	$\cdot$	10^{-1}	10^{-2}	10^{-3}	10^{-4}	$\cdots$

Example:

(i) $7693 = 7 \times 10^3 + 6 \times 10^2 + 9 \times 10^1 + 3 \times 10^0$

$= 7 \times 1000 + 6 \times 100 + 9 \times 10 + 3 \times 1$

$= 7000 + 600 + 90 + 3$

(ii) $1936.46 = 1 \times 10^3 + 9 \times 10^2 + 3 \times 10^1 + 6 \times 10^0 + 4 \times 10^{-1} + 6 \times 10^{-2}$

$= 1000 + 900 + 30 + 6 + 0.4 + 0.06$

1.3 Binary Number System:

The base of the binary number system is two. It uses the digits0 and 1 only. The two digits 0 and 1 are called a bit. The place value of each position can be expressed in terms of powers of 2 like $2^0, 2^1, 2^2, etc$ for integer part and $2^{-1}, 2^{-2}, 2^{-3}, etc$ for the fractional part. A binary point separates the integer and fractional part. The position weights in the binary is given as

$\cdots$	2^3	2^2	2^1	2^0	$\cdot$	2^{-1}	2^{-2}	2^{-3}	2^{-4}	$\cdots$

$4\ bit\ binary\ word \Longrightarrow \quad nibble$

$8\ bit\ binary\ word \Longrightarrow \quad byte$

$16\ bit\ binary\ word \Longrightarrow \quad word$

$32\ bit\ binary\ word \Longrightarrow \quad double\ word$

1.4 Octal Number System:

The base of the octal number system is eight. It uses eight digits $0\ 1\ 2\ 3\ 4\ 5\ 6\ and\ 7$ to form a number. The place value of each position can be expressed in terms of powers of 8 like $8^0, 8^1, 8^2, etc$ for integer part and $8^{-1}, 8^{-2}, 8^{-3}, etc$ for the fractional part. An octal point separates the integer and fractional part. Sets of 3-bit binary numbers can be represented by octal numbers $(000, 001, 010, 011, 100, 101, 110, 111)$ and this can be conveniently be used for entering data in the computer. The position weights in the octal system is given as

$\cdots$	8^3	8^2	8^1	8^0	$\cdot$	8^{-1}	8^{-2}	8^{-3}	8^{-4}	$\cdots$

1.5 Hexadecimal Number System:

The Hexadecimal number system has a base of 16. It has 16 distinct digit symbols. It uses the digits $0\ 1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9$ plus the letters $A\ B\ C\ D\ E\ and\ F$. Any hexadecimal digit can be represented by a group of four bit binary sequence.That is the Hexadecimal numbersare represented by sets of 4-bit binary sequence (0000, 0001,0010, 0011, 0100,0101,0110, 0111,1000,1001,1010,1011,1100,1101,1110,1111). The position weight in the hexadecimal number system is given as

$\cdots$	16^3	16^2	16^1	16^0	$\cdot$	16^{-1}	16^{-2}	16^{-3}	16^{-4}	$\cdots$

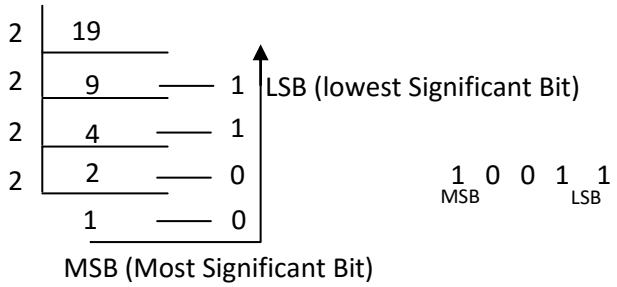
Number System

Decimal (Base 10)	**Binary (Base 2)**	**Octal (Base 8)**	**Hexadecimal (Base 16)**
0	0000	00	0
1	0001	01	1
2	0010	02	2
3	0011	03	3
4	0100	04	4
5	0101	05	5
6	0110	06	6
7	0111	07	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

1.6 Decimal to Binary Conversion:

Decimal number can be convertedto binary by repeatedly dividing the number by 2 for integer part and collecting the reminders.The remainders can be written in the reverse order(from bottom to top) to get binary result. For fractional part, it has to be multiplied by 2 successively and collecting the carries, to write from top to bottom. The multiplication is repeated till the fractional part becomes zero or the required number of significant bit is obtained.

1. Convert $(19)_{10}$ into its Binary equivalent



$$(19)_{10} = (10011)_2$$

2. Convert$(0.625)_{10}$ into its Binary equivalent

$0.625 \times 2 = 1.250$ $carry\ is$ 1 (MSB)

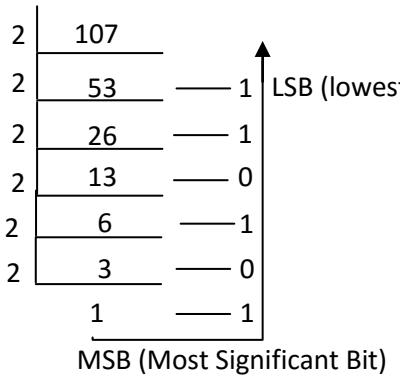
$0.250 \times 2 = 0.500$ $carry\ is$ 0

$0.500 \times 2 = 1.000$ $carry\ is$ 1 (LSB)

1 0 1
MSB LSB

$$(0.625)_{10} = (0.101)_2$$

3. Convert $(107.6875)_{10}$ to its equivalent Binary



$0.6875 \times 2 = 1.3750$ $carry\ is$ 1 (MSB)

$0.3750 \times 2 = 0.7500$ $carry\ is$ 0

$0.7500 \times 2 = 1.500$ $carry\ is$ 1

$0.5000 \times 2 = 1.000$ $carry\ is$ 1 (LSB)

1 0 1 1
MSB LSB

$$(107.6875)_{10} = (1101011.1011)_2$$

1.7 Binary to Decimal Conversion:

A binary number can be converted into decimal number by adding the products of eachbitanditscorresponding weight.

Example:

(i) $(101)_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 4 + 0 + 1$

$= (5)_{10}$

(ii) $(10011)_2 = 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$= 16 + 0 + 0 + 2 + 1$

$= (19)_{10}$

(iii) $(0.101)_2 = 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$

$= 1 \times 0.5 + 0 + 1 \times 0.125$

$= 0.5 + 0 + 0.125$

$= (0.625)_{10}$

(iv) $(1101011.1011)_2$

$(1101011)_2 = 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^1$

$= 64 + 32 + 0 + 8 + 0 + 2 + 1$

$= (107)_{10}$

$(0.1011) \quad = 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}$

$= 0.5 + 0 + 0.125 + 0.0625$

$= (0.6875)_{10}$

$\therefore (1101011.1011)_2 = (107.6875)_{10}$

1.8 Hexadecimal to Decimal:

A hexadecimal number can be converted into decimal number by adding the products of each digit and its corresponding weight. The weights are power of 16.

Example:

1. Convert hexadecimal $(D5)_{16}$ to decimal

$(D5)_{16} = (13 \times 16^1 + 5 \times 16^0) \Longrightarrow \ (D)_{16} = (13)_{10}$

$= 13 \times 16 + 5 \times 1$

$= 208 + 5$

$= (213)_{10}$

$ie., (D5)_{16} = (213)_{10}$

2. Convert hexadecimal $(3FC.8)_{16}$ to decimal

$$(3FC.8)_{16} = (3 \times 16^2 + 15 \times 16^1 + 12 \times 16^0 + 8 \times 16^{-1})$$

$$\Longrightarrow (F)_{16} = (15)_{10} \& (C)_{16} = (12)_{10}$$

$$= 3 \times 256 + 15 \times 16 + 12 \times 1 + 8 \times \frac{1}{16}$$

$$= 768 + 240 + 12 + 0.5$$

$$= (1020.5)_{10}$$

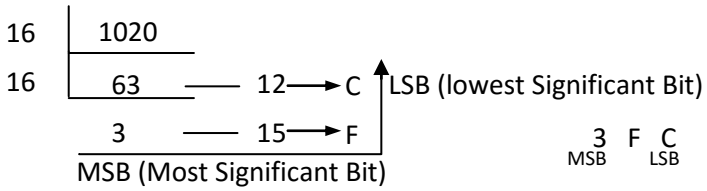
$$ie., (3FC.8)_{16} = (1020.5)_{10}$$

1.9 Decimal to Hexadecimal:

Decimal number can be converted to hexadecimal by repeatedly dividing the number by 16 for integer part and collecting the reminders. The remainders can be written in the reverse order (from bottom to top) to get hex result. For fractional part, it has to be multiplied by 16 successively and collecting the carries, to write from top to bottom. The multiplication is repeated till the fractional part becomes zero or the required numbers of significant digits are obtained.

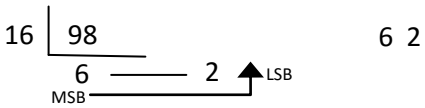
Example:

1. Convert $(1020)_{10}$ to hexadecimal



$$(1020)_{10} = (3FC)_{16}$$

2. Convert $(98.625)_{10}$ to hexadecimal



$$0.625 \times 16 = 10.000 \quad carry\ is\ \ 10 \Longrightarrow A$$

$$(98.625)_{10} = \ (62.A)_{16}$$

1.10 HexadecimaltoBinary:

Hexadecimal numbers can be converted into binary numbers by converting each hexadecimal digit to its 4-bit binary equivalent

1. Convert $(25)_H$ to Binary

   $(25)_H = \left(\underbrace{0010}_{2}\underbrace{0101}_{5}\right)_2$

2. Convert $(3A.7)_H$ to Binary

   $(3A)_H = \left(\underbrace{0011}_{3}\underbrace{1010}_{A}\right)_2$

   $(.7)_H = \left(\underbrace{.0111}_{7}\right)_2$

   $(3A.7)_H = \left(\underbrace{0011}_{3}\underbrace{1010}_{A} \cdot \underbrace{0111}_{7}\right)_2$

3. Convert $(CD.E8)_H$ to Binary

   $(CD)_H = \left(\underbrace{1100}_{C}\underbrace{1101}_{D}\right)_2$

   $(.E8)_H = \left(\underbrace{.1110}_{E}\underbrace{1000}_{8}\right)_2$

   $(CD.E8)_H = \left(\underbrace{1100}_{C}\underbrace{1101}_{D} \cdot \underbrace{1110}_{E}\underbrace{1000}_{8}\right)_2$

1.11 Binary to Hexadecimal:

Conversion from binary to hexadecimal is easily accomplished by partitioning the binary number into groups of four binary digits, starting from the binary point to the left and to the right. It may be necessary to add zeros to the last group, if it does not end in exactly four bits. Each group of 4-bits binary must be represented by its hexadecimal equivalent.

1. $(1010 \cdot 1101)_2 = (A \cdot D)_H$
2. $(110 \cdot 101)_2 = (0110 \cdot 1010)_2 = (6 \cdot A)_H$
3. $(1110 \cdot 11)_2 = (1110 \cdot 1100)_2 = (E \cdot C)_H$

1.12 Octal to Decimal:

Anoctal number can be converted into decimal number by adding the products of each digit and its corresponding weight. The weights are power of 8.

1. $(75)_8 = 7 \times 8^1 + 5 \times 8^0$

   $= 56 + 5$

   $= (61)_{10}$

2. $(45 \cdot 6)_8 = 4 \times 8^1 + 5 \times 8^0 + 6 \times 8^{-1}$

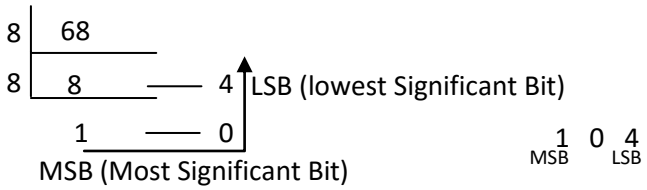
   $= 32 + 5 + 0.75$

   $= (37.75)_{10}$

1.13 Decimal to octal:

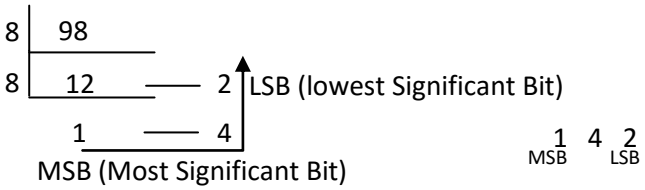
Decimal number can be converted to octal by repeatedly dividing the number by 8 for integer part and collecting the reminders. The remainders can be written in the reverse order (from bottom to top) to get octal result. For fractional part, it has to be multiplied by 8 successively and collecting the carries, to write from top to bottom. The multiplication is repeated till the fractional part becomes zero or the required numbers of significant digits are obtained.

1. Convert $(68)_{10}$ to octal



$$(68)_{10} = (104)_8$$

2. Convert $(98.625)_{10}$to Octal



$$0.625 \times 8 = 5.0000 \ \ carry\ is\ \ 5$$

$$(98.625)_{10} = (142.5)_8$$

1.14 Octal to Binary:

Octal numbers can be converted into binary numbers by converting each octal digit to its 3-bit binary equivalent

1. $(27)_8 = \left(\underbrace{010}_{2}\ \underbrace{111}_{7}\right)_2$
2. $(135)_8 = \left(\underbrace{001}_{1}\ \underbrace{011}_{3}\ \underbrace{101}_{5}\right)_2$
3. $(45.5)_8 = \left(\underbrace{100}_{4}\ \underbrace{101}_{5} \cdot \underbrace{101}_{5}\right)_2$

1.15 Binary to Octal:

Conversion from binary to octal is the simplest procedure by grouping the binary number into groups of three binary digits, starting from the binary point to the left and to the right. It may be necessary to add zeros to the last group, if it does not end in exactly three bits. Each group of 3-bits binary must be represented by its octal equivalent.

1. $(10\ 101)_2 = (010\ 101)_2 = (25)_8$
2. $(101011)_2 = (101\ 011)_2 = (53)_8$
3. $(11110.11)_2 = (011\ 110\ .\ 110)_2 = (36.6)_8$
4. $(11011011.1111)_2 = (011\ 011\ 011.\ 111\ 100)_2 = (333.74)_8$

1.16 Hexadecimal to Octal:

This can be achieved by first writing down the four bit binary equivalent of hexadecimal digit and then partitioning it into group of 3 bits each. Finally, the three bit octal equivalent is written down.

Example:

1. Convert $(2AB.9)_H$ to octal

$$\begin{array}{lcccccccc} Hexa\ decimal\ Number & \rightarrow & 2 & & A & & B & . & 9 \\ & & \downarrow & & \downarrow & & \downarrow & . & \downarrow \\ 4\ bit\ Binary\ Number & \rightarrow & 0010 & & 1010 & & 1011 & . & 1001 \\ 3\ bit\ Partition & \rightarrow & \underbrace{001}\ \underbrace{010}\ \underbrace{101}\ \underbrace{011} & & & & & . & \underbrace{100}\ \underbrace{100} \\ & & \downarrow & \downarrow & \downarrow & \downarrow & & . & \downarrow\ \downarrow \\ Octal\ Number & \rightarrow & 1 & 2 & 5 & 3 & . & 4 & 4 \end{array}$$

$$\therefore (2AB.9)_{16} = (1253.44)_8$$

2. Convert $(3FC.82)_H$ to octal

$$\begin{array}{lcccccccc} Hexa\ decimal\ Number & \rightarrow & 3 & & F & & C & . & 8\quad 2 \\ & & \downarrow & & \downarrow & & \downarrow & . & \downarrow\quad \downarrow \\ 4\ bit\ Binary\ Number & \rightarrow & 0011 & & 1111 & & 1100 & . & 1000\quad 0010 \\ 3\ bit\ Partition & \rightarrow & \underbrace{001}\ \underbrace{111}\ \underbrace{111}\ \underbrace{100} & & & & & . & \underbrace{100}\ \underbrace{000}\ \underbrace{100} \\ & & \downarrow & \downarrow & \downarrow & \downarrow & & . & \downarrow\ \downarrow\ \downarrow \\ Octal\ Number & \rightarrow & 1 & 7 & 7 & 4 & . & 4 & 0\quad 4 \end{array}$$

$$\therefore (3FC.82)_{16} = (1253.44)_8$$

1.17 Octal to Hexadecimal:

This can be achieved by first writing down the three bit binary equivalent of octal digit and then partitioning it into group of 4 bits each. Finally, the four bit hexadecimal equivalent is written down.

1. Convert $(16.2)_8$ to Hexadecimal

$$
\begin{array}{lcccccc}
octal\ Number & \rightarrow & 1 & 6 & . & 2 \\
 & & \downarrow & \downarrow & . & \downarrow \\
3\ bit\ Binary & \rightarrow & 001 & 110 & . & 010 \\
 & & \downarrow & \downarrow & . & \downarrow \\
4\ bit\ Partition & \rightarrow & \underbrace{0000}\underbrace{1110} & . & \underbrace{0100} \\
 & & \downarrow & \downarrow & . & \downarrow \\
Hex\ Number & \rightarrow & 0 & E & . & 4
\end{array}
$$

$$\therefore (16.2)_8 = (0E.4)_{16} = (E.4)_H$$

A zero is added to the right most group to make it a group of 4 bits and left most zeros are dropped

2. Convert $(764.352)_8$ to Hexadecimal

$$
\begin{array}{lcccccccccc}
octal\ Number & \rightarrow & 7 & 6 & 4 & . & 3 & 5 & 2 \\
 & & \downarrow & \downarrow & \downarrow & . & \downarrow & \downarrow & \downarrow \\
3\ bit\ Binary & \rightarrow & 111 & 110 & 100 & . & 011 & 101 & 010 \\
 & & \downarrow & \downarrow & \downarrow & . & \downarrow & \downarrow & \downarrow \\
4\ bit\ Partition & \rightarrow & \underbrace{0001}\underbrace{1111}\underbrace{0100} & . & \underbrace{0111}\underbrace{0101}\underbrace{0000} \\
 & & \downarrow & \downarrow & \downarrow & . & \downarrow & \downarrow & \downarrow \\
Hex\ Number & \rightarrow & 1 & F & 4 & . & 7 & 5 & 0
\end{array}
$$

$$\therefore (764.352)_8 = (1F4.750)_{16} = (1F4.75)_H$$

1.18 Binary Arithmetic:

Arithmetic operations such as addition, subtraction, multiplication and division can be performed on binary numbers.

1.18.1 Binary addition:

The addition of two Binary numbers is very similar to addition of two decimal numbers. It is key to binary subtraction, multiplication and division.The following rules are followed while adding two binary numbers.